

Kepler-17b

R-2404

Benchmark run · workflow W8-benchmark-deep · record deep-bench_KEPLER-17_180630 ·
created 2026-08-21T00:44:36+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458299.73044 +/- 0.00084 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.18 +/- 0.03
Transit depth (Rp/Rs)^2	3.25 +/- 1.07 [%]
Orbital Inclination (inc)	89.45 +/- 1.58
Airmass coefficient 1 (a1)	0.975 +/- 0.084
Airmass coefficient 2 (a2)	0.004 +/- 0.035
Scatter in the residuals of the lightcurve fit is	8.59 %
Best Comparison Star	#3 - [503, 241]
Optimal Aperture	2.12
Optimal Annulus	4.03
Transit Duration (day)	0.099 +/- 0.0029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.18 ± 0.03 vs 0.13031 (deviation 1.11σ)
Duration fit vs published	0.099 d vs 0.0954 d (deviation 0.83σ)
Mid-time O–C	2.1 min
Depth SNR	4.0
Residual scatter	8.59 %

Light curve

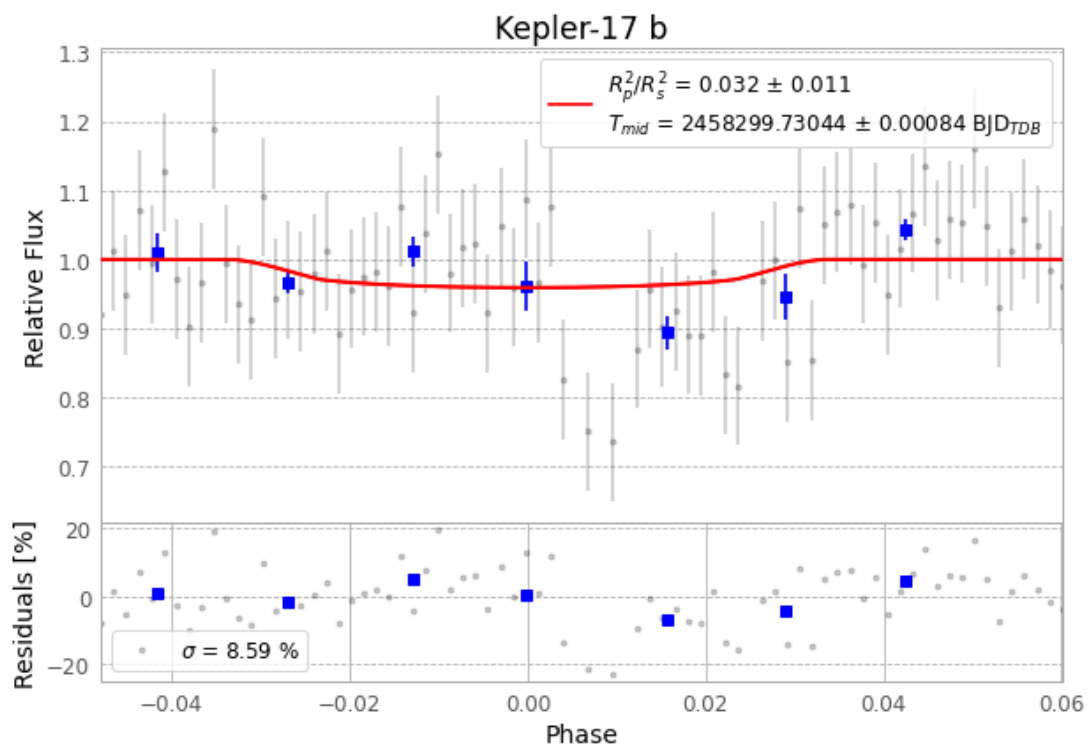


Figure 1 — FinalLightCurve_Kepler-17 b_2018-06-29.png (SHA-256 c00f4d11de202ba9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [386, 196], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 395a5e106aebc65b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

78 FITS frames (51 MB), KEPLER-17180630034512.FITS → KEPLER-17180630073612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-KEPLER-17-180630.log (4053dcdce2cc...), FinalLightCurve_Kepler-17 b_2018-06-29.png (c00f4d11de20...), FinalLightCurve_Kepler-17 b_2018-06-29.csv (de04d25edcc2...)

Compute resources

Wall time 107.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-21 00:44Z.